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MCY 611
Lab 6 AWS CloudFront

MCY 611 Cloud Computing
Lab#6: Amazon CloudFront

Part 1: Create CloudFront distribution

1.Make a bucket via the following mb command:
aws s3 mb s3://*your_user_name*
What is the output?

```
~ $ aws s3 mb s3://carrd12
make_bucket: carrd12
```

2.Delete public access block.
aws s3api delete-public-access-block --bucket *your_user_name*

3.Set a bucket policy to allow public read access. First, create a file called policy.json with the following contents:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::your_user_name/*"
            ]
        }
    ]
}
```

Set the bucket policy with this command:
aws s3api put-bucket-policy --bucket *your_user_name* --policy file://policy.json
What is your command and its output?

```
~ $ aws s3api put-bucket-policy --bucket carrd12 --policy file://policy.json
```

4.Download our MSCYS program webpage via the wget command, which downloads files from the Internet.

wget https://onlinedegrees.nku.edu/programs/technology/ms-cybersecurity/

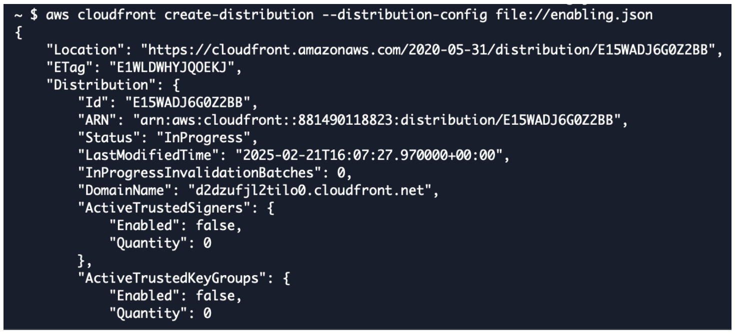
An index.html file should be saved under your current directory.
Copy index.html to your bucket via the following command:
aws s3 cp index.html s3://*your_user_name*/index.html
What is your command and its output?

```
~ $ aws s3 cp index.html s3://carrd12/index.html
upload: ./index.html to s3://carrd12/index.html
```

5.Create a distribution configuration json file named enabling.json in the current folder. It defines a CloudFront distribution. You can use vi to create the json file. Once vi is open, press i, it will change to the insertion mode. Then copy the content below (you need to replace words in italics with your information) and paste it into the terminal. Then press ESC, and type :wq.

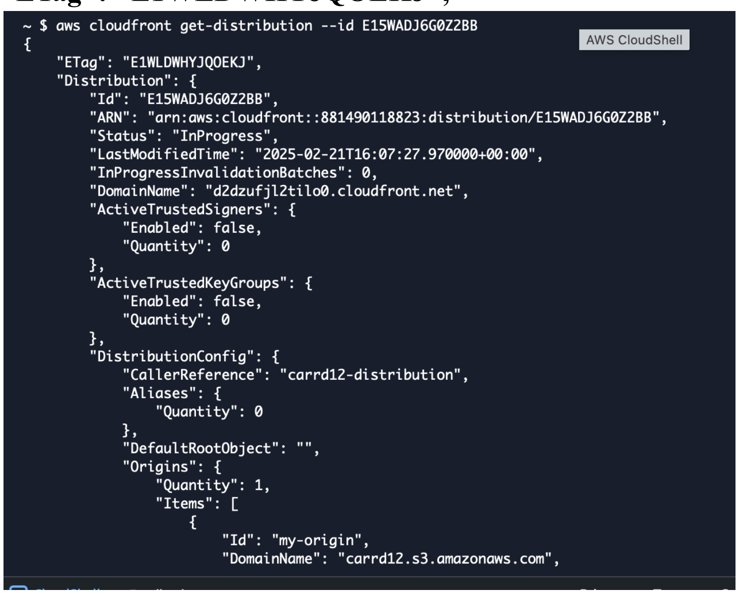
```
{
 "CallerReference": "your_user_name-distribution",
 "Aliases": {
  "Quantity": 0
 },
 "DefaultRootObject": "",
 "Origins": {
  "Quantity": 1,
  "Items": [
   {
    "Id": "my-origin",
    "DomainName": "your_user_name.s3.amazonaws.com",
    "S3OriginConfig": {
     "OriginAccessIdentity": ""
    }
   }
  ]
 },
 "DefaultCacheBehavior": {
  "TargetOriginId": "my-origin",
  "ForwardedValues": {
   "QueryString": true,
   "Cookies": {
    "Forward": "none"
   }
  },
  "TrustedSigners": {
   "Enabled": false,
   "Quantity": 0
  },
  "ViewerProtocolPolicy": "allow-all",
  "MinTTL": 3600
 },
 "CacheBehaviors": {
  "Quantity": 0
 },
 "Comment": "",
 "Logging": {
  "Enabled": false,
  "IncludeCookies": true,
  "Bucket": "",
  "Prefix": ""
 },
 "PriceClass": "PriceClass_All",
 "Enabled": true
}
```

Create the distribution via the following command:
aws cloudfront create-distribution --distribution-config file://enabling.json
What is the output? What is your distribution ID? Here is an example of a distribution ID in my output: "Id": "E3I7WEK9ENIFDF"
"Id": "E15WADJ6G0Z2BB",



```
~ $ aws cloudfront create-distribution --distribution-config file://enabling.json
{
    "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E15WADJ6G0Z2BB",
    "ETag": "E1WLDWHYJQOEKJ",
    "Distribution": {
        "Id": "E15WADJ6G0Z2BB",
        "ARN": "arn:aws:cloudfront::881490118823:distribution/E15WADJ6G0Z2BB",
        "Status": "InProgress",
        "LastModifiedTime": "2025-02-21T16:07:27.970000+00:00",
        "InProgressInvalidationBatches": 0,
        "DomainName": "d2dzufjl2tilo0.cloudfront.net",
        "ActiveTrustedSigners": {
            "Enabled": false,
            "Quantity": 0
        },
        "ActiveTrustedKeyGroups": {
            "Enabled": false,
            "Quantity": 0
```

6.Check the information about a distribution via the following command:
aws cloudfront get-distribution --id *your_distribution_id*
What is the output? What are the values of "Status" and "ETag" in the output?
"Status": "InProgress",
"ETag": "E1WLDWHYJQOEKJ",




```
~ $ aws cloudfront get-distribution --id E15WADJ6G0Z2BB
{
    "ETag": "E1WLDWHYJQOEKJ",
    "Distribution": {
        "Id": "E15WADJ6G0Z2BB",
        "ARN": "arn:aws:cloudfront::881490118823:distribution/E15WADJ6G0Z2BB",
        "Status": "InProgress",
        "LastModifiedTime": "2025-02-21T16:07:27.970000+00:00",
        "InProgressInvalidationBatches": 0,
        "DomainName": "d2dzufjl2tilo0.cloudfront.net",
        "ActiveTrustedSigners": {
            "Enabled": false,
            "Quantity": 0
        },
        "ActiveTrustedKeyGroups": {
            "Enabled": false,
            "Quantity": 0
        },
        "DistributionConfig": {
            "CallerReference": "carrd12-distribution",
            "Aliases": {
                "Quantity": 0
            },
            "DefaultRootObject": "",
            "Origins": {
                "Quantity": 1,
                "Items": [
                    {
                        "Id": "my-origin",
                        "DomainName": "carrd12.s3.amazonaws.com",
```

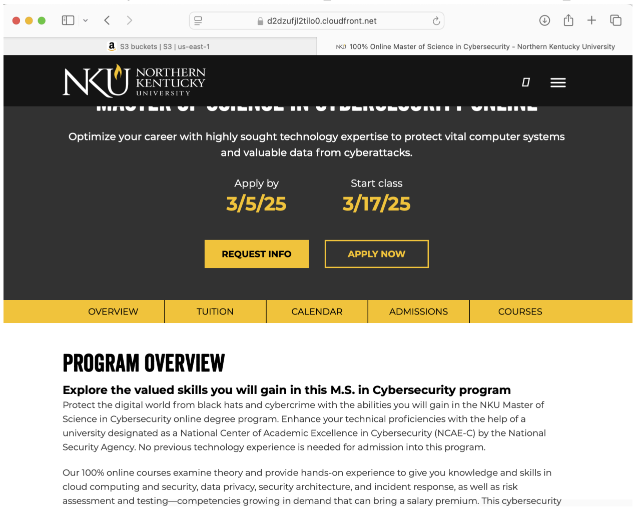
An S3 bucket can be used as the origin of an Amazon CloudFront distribution. The S3 bucket's domain name is inside the Origins section of the output. CloudFront uses this to fetch content from S3. **What is your S3 bucket's domain name?**
https://carrd12.s3.amazonaws.com

The CloudFront domain name is found under the DomainName field in the Distribution section of the output. It should end with cloudfront.net. This CloudFront domain name is the URL used to access content served via CloudFront. **What is your CloudFront domain name?**
d2dzufjl2tilo0.cloudfront.net

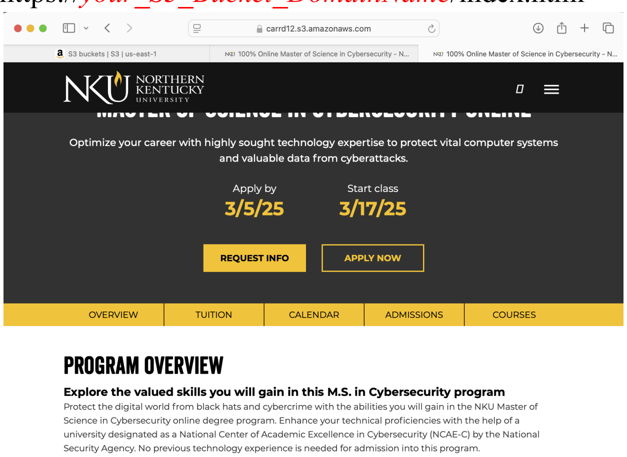
If the status is "InProgress", then you have to wait. The status will turn from "InProgress" to "Deployed" when the distribution is successfully created. It will take about 10 minutes to see the status changes to "Deployed". You can run the get-distribution command again in 10 minutes.

7.When the status turns to "Deployed", open a browser and visit your CloudFront URL:
https://*your_CloudFront_DomainName*/index.html

What did you see? Please provide a screen capture.



Visit your S3 URL:
https://*your_S3_Bucket_DomainName*/index.html



Part 2: Performance testing

8.You will use the Pingdom site to compare the performance between the S3(origin) and its CloudFront distribution. Go to http://tools.pingdom.com/fpt/ Test the load time of the MSCYS program page at the S3 bucket and its CloudFront distribution. At step 7, you got the s3 URL and CloudFront URL of the MSCYS program page. The Pingdom site can make web requests from multiple locations. Test each URL from two different locations. Repeat each URL test twice. Your CloudFront distribution does not copy files to all edge locations immediately. The first access to CloudFront triggers a copy of the data to the edge location, which will then cache the content for future accesses, so the second access will have a lower latency than the first access.

Note: after each test, make sure to clear your browser cache before starting the next test. Additionally, be aware that Pingdom may display a message stating, "You've reached your limit". If this happens, wait for some time before attempting the test again.

Fill in the table below with the data from your tests.

URL	Test From	Load Time
https://carrd12.s3.amazonaws.com/index.html	Tokyo, Japan	2.09s
https://carrd12.s3.amazonaws.com/index.html	Tokyo, Japan	1.48s
https://carrd12.s3.amazonaws.com/index.html	Frankfurt, Germany	1.33s
https://carrd12.s3.amazonaws.com/index.html	Frankfurt, Germany	1.03s
https://d2dzufjl2tilo0.cloudfront.net/index.html	Tokyo, Japan	2.41s
https://d2dzufjl2tilo0.cloudfront.net/index.html	Tokyo, Japan	1.09s
https://d2dzufjl2tilo0.cloudfront.net/index.html	Frankfurt, Germany	2.06s
https://d2dzufjl2tilo0.cloudfront.net/index.html	Frankfurt, Germany	831ms

How do the latencies of these two URLs compare? Which is faster? Is it the one you expected?
Looking over the data from my test both S3 and CF urls load faster in Frankfurt compared to Tokyo, the S3 hosted site is slightly more consistent across tests averaging 1.79s in Tokyo and 1.18s in Frankfurt, and CF holds the fastest time at 831ms out of Frankfurt but also the slowest at 2.41s out of Tokyo.
The CF URL loads in 1.60 seconds on average, whereas the S3-hosted URL loads in 1.48 seconds on average. This indicates that, on average, S3 performs slightly faster than CF in this test; this is probably because CF has network routing problems or cache misses.
I expected the CF to load faster.

Part 3: Cleanup
9.Disable your distribution. Generate a file named disabling.json via the following:
aws cloudfront get-distribution-config --id *your_distribution_id* | jq '. | .DistributionConfig' > disabling.json

Open the disabling.json file by typing vi disabling.json. Find the line, "Enabled": true,, below the "PriceClass": "PriceClass_All" and change true to false, by typing i and then changing to false. Save the change, press ESC, then type :wq. The jq command is used to transform JSON data into a more readable format. It is built around filters which are used to find and print only the required data from a JSON file.
Run the following command to disable your distribution:
aws cloudfront update-distribution --id *your_distribution_id* --distribution-config file://disabling.json --if-match *your_etag*

You can find *your_distribution_id* and *your_etag* at step 5 and 6.
What is your command and its output?

```
~ $ aws cloudfront update-distribution --id E15WADJ6G0Z2BB --distribution-config file://disabling
.json --if-match E1WLDWHYJQOEKJ
{
    "ETag": "E5LE0539BUNNR",
    "Distribution": {
        "Id": "E15WADJ6G0Z2BB",
        "ARN": "arn:aws:cloudfront::881490118823:distribution/E15WADJ6G0Z2BB",
        "Status": "InProgress",
        "LastModifiedTime": "2025-02-21T17:06:28.284000+00:00",
        "InProgressInvalidationBatches": 0,
        "DomainName": "d2dzufjl2tilo0.cloudfront.net",
        "ActiveTrustedSigners": {
            "Enabled": false,
            "Quantity": 0
        },
        "ActiveTrustedKeyGroups": {
            "Enabled": false,
            "Quantity": 0
        },
        "DistributionConfig": {
            "CallerReference": "carrd12-distribution",
            "Aliases": {
                "Quantity": 0
```

10.Check the information about a distribution via the following command:
aws cloudfront get-distribution --id *your_distribution_id*
If the status is "InProgress", then you have to wait. The status will turn from "InProgress" to "Deployed" when the distribution is successfully disabled. It will take about 10 minutes to see the status changes to "Deployed". You can run the get-distribution command again in 10 minutes.

11.When the status turns to "Deployed", **what is your ETag value in the output of the get-distribution command?** You need this value in the command below. You can delete the distribution via the following:
aws cloudfront delete-distribution --id *your_distribution_id* --if-match *your_etag*
What is your command?
"ETag": "E5LE0539BUNNR",

```
~ $ aws cloudfront delete-distribution --id E15WADJ6G0Z2BB --if-match E5LE0539BUNNR
```

12.Please remove your s3 bucket. **What is your command and its output?**

```
~ $ aws s3 rb s3://carrd12 --force
delete: s3://carrd12/index.html
remove_bucket: carrd12
```

Submission instruction
After completing this work, submit this word document with your answers to the Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

Studying Amazon CloudFront, AWS's content delivery network (CDN), and how it enhances web content distribution's scalability, security, and performance were the objectives of this lab. I set up caching behaviors, controlled CloudFront distributions, and kept an eye on performance using the AWS CLI. In order to accomplish these objectives, I first used the AWS CLI to build a CloudFront distribution, indicating the origin (such as an EC2 instance or S3 bucket) where the material is kept. After that, I set up cache behaviors, specifying how CloudFront ought to respond to various requests and kinds of content. I then turned on HTTPS and security settings like limiting origins and limiting access via signed URLs. I tested the distribution by going to the CloudFront URL and looking at latency improvements once I had set it up. Lastly, I tracked request trends and enhanced performance using AWS CloudWatch and CloudFront logs.